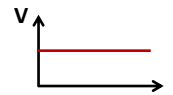
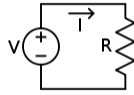
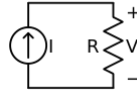


Voltage and current sources, voltage and current meters

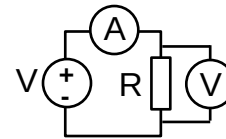
Ideal voltage generator: a voltage source whose output voltage does not depend on the resistance (or current) of the load circuit



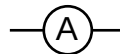
Ideal current generator: a current source whose current does not depend on the resistance (or voltage) of the load circuit



Ideal voltmeter: voltmeter with infinite internal resistance, connected in parallel with the circuit element to be measured. The presence of the voltmeter does not affect the current flowing through the measured object!



Ideal current meter: a current meter with zero internal resistance, connected in series with the circuit element to be measured. The presence of the current meter does not affect the voltage on the measured object!

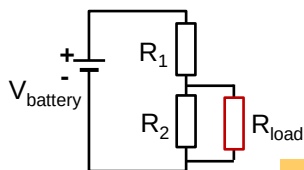


On most power supplies both voltage and current can be set. From the both, the more restricting one is realized. Example: 30V & 10A is set at 100Ω load. Current of 10A would require 1000V, i.e. 30V is realized on the output at 0.3A current, and the CV light is on (constant voltage). Example 2: at 1Ω resistor 10A and 10V is realized (CC-constant current light is on).

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Simple voltage and current generator circuit

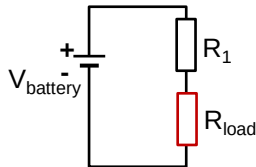
A battery and a voltage divider as a voltage source:



$$V_{\text{load}} = V_{\text{battery}} \cdot \frac{R_2}{R_1 + R_2}, \text{ if } R_{\text{load}} \gg R_2 = R_{\text{output}}$$

Output resistance of the voltage source (ideally zero).
The desired voltage is only obtained, if the load resistance is much larger than the output resistance.

A battery and a resistor as a current source:



$$I_{\text{load}} = \frac{V_{\text{battery}}}{R_1}, \text{ if } R_{\text{load}} \ll R_1 = R_{\text{output}}$$

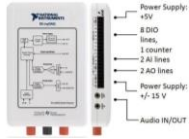
Output resistance of the current source (ideally infinity). The desired current is only obtained if the load resistance is much smaller than the output resistance.

Notes: such battery circuits are not only simplified examples, but a battery is also an ideal choice if a source with very low internal noise level is desired. The disadvantage is that the voltage of the battery is not adjustable. As a further note, the above voltage divider circuit is also useful, if an adjustable voltage source is used instead of a battery, as the voltage division also divides the output noise level (i.e. for 1mV output it might be better to output 100 mV with 1:100 division, as the direct 1mV setting of the voltage source may have larger output noise level).



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A data acquisition (DAQ) card is a general-purpose instrument which acts both as a versatile voltmeter and a programmable voltage source:



Inputting and outputting arbitrary voltage patterns

The operation of general digital voltage sources and voltmeters relies on analog-to-digital (A/D) and digital-to-analog (D/A) converters. These devices include several amplifiers. Though the operation of amplifiers is a subject of electronics courses, here a brief insight is given to A/D and D/A converters as well as the basics of amplifier circuits.

Basics of operational amplifiers (op-amps) treated as a black-box:

The op-amp is an integrated circuit with multiple pins:

- One positive and one negative supply voltage (e.g. +15V and -15V): V_{S+} , V_{S-}
- Two input voltages (a non-inverting and an inverting one): V_{IN+} , V_{IN-}
- An output voltage: V_{OUT}

The **amplified input voltage difference is outputted**: $V_{OUT} = A_{OL} \cdot (V_{IN+} - V_{IN-})$, where A_{OL} is the so-called **open loop voltage gain**. A_{OL} is usually very large (e.g. $A_{OL} \approx 10^5$). If the output voltage would approach the positive/negative supply voltage, rather a so-called saturation voltage is outputted (V_{sat+} , V_{sat-}), which is close to the positive/negative supply voltage, see red curve.

Practically **no current flows at the two inputs** (V_{IN+} , V_{IN-}), i.e. the input resistance is practically infinite.

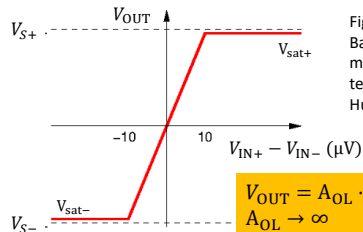
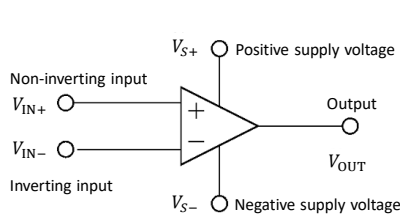


Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

$$V_{OUT} = A_{OL} \cdot (V_{IN+} - V_{IN-})$$

$$A_{OL} \rightarrow \infty$$

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Using an op-amp as a comparator

Simplest application: comparator circuit. A reference voltage V_{ref} is connected to the negative input, and the investigated input voltage V_{IN} is connected to the positive input. Due to the close-to-infinity (very large) open loop gain already a relatively small voltage difference compared to V_{ref} yields a saturated voltage at the output such that:

- $V_{OUT} = V_{sat-}$ if $V_{IN} < V_{ref}$
- $V_{OUT} = V_{sat+}$ if $V_{IN} > V_{ref}$

Therefore, the output voltage can be used to decide whether the V_{IN} input signal is greater or less than the V_{ref} reference signal (the input is *compared* to the reference -> comparator).

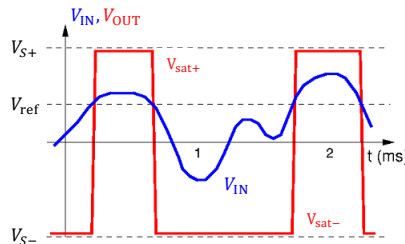
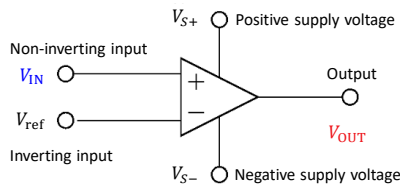


Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

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Using the op-amp as a voltage amplifier (non-inverting voltage amplifier circuit)

The output of the amplifier is fed back to the negative input through a voltage divider with R_1 and R_2 resistances (**negative feedback**):

$$V_{IN-} = \frac{R_2}{R_1 + R_2} V_{OUT}, \text{ furthermore, } V_{OUT} = A_{OL} \cdot (V_{IN+} - V_{IN-})$$

$$\text{From these: } \frac{R_1 + R_2}{R_2} V_{IN-} = A_{OL} \cdot (V_{IN+} - V_{IN-})$$

From these, also considering the $A_{OL} \rightarrow \infty$ condition:

$$V_{IN-} = V_{IN+} \cdot \frac{R_2 A_{OL}}{R_2 A_{OL} + R_1 + R_2} \approx V_{IN+}, \text{ i.e. due to the negative feedback the potentials of the „+” és „-” inputs are the same: } V_{IN} = V_{IN+} = V_{IN-}.$$

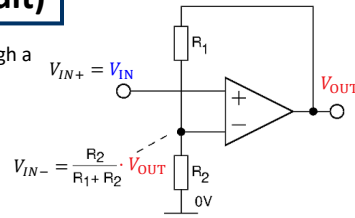


Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

From all these

$$V_{OUT} = \frac{R_1 + R_2}{R_2} V_{IN}$$

follows, i.e. the proper voltage amplification can be set by the R_1 and R_2 resistances

General rules for calculating amplifier circuits:

- (i) The current is zero at the two inputs (i.e. both the inverting and non-inverting input)
- (ii) The inverting and non-inverting inputs have the same potential ($V_{IN+} = V_{IN-}$)

Note, that rule (ii) is not general for all amplifier circuits, it only applies for cases with negative feedback (i.e. the output is fed back to the negative input). I.e. rule (ii) does not apply for a comparator circuit, but it applies to most voltage and current amplifier circuits, as all these rely on negative feedback.

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Using the op-amp as a voltage amplifier (inverting voltage amplifier circuit)

The positive input is grounded ($V_{IN+} = 0$) and the output is fed back to the negative input through the R_1 resistor. Due to rule (ii) the negative input is also at virtual ground, $V_{IN-} = 0$. The current on the R_2 resistor is calculated as:

$$I_{R_2} = \frac{V_{IN} - V_{IN-}}{R_2} = \frac{V_{IN}}{R_2}$$

Furthermore, the I_{R_2} current flowing through R_2 cannot flow into the negative input of the amplifier (see rule (i)), i.e. the entire I_{R_2} current flows through the R_1 resistor as well ($I_{R_2} = I_{R_1}$):

$$I_{R_1} = \frac{V_{IN-} - V_{OUT}}{R_1} = -\frac{V_{OUT}}{R_1} = I_{R_2} = \frac{V_{IN}}{R_2}$$

From the $I_{R_2} = I_{R_1}$ condition

$$V_{OUT} = -\frac{R_1}{R_2} V_{IN}$$

follows, i.e. the proper voltage amplification can be set by the R_1 and R_2 resistance ratio.

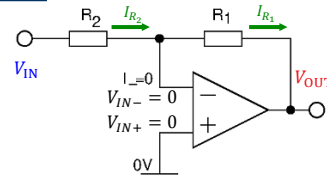
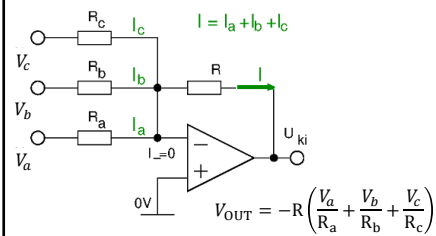


Figure source: Varga-Bagoly: Electronics and measurement techniques (in Hungarian)

Generalization: summing amplifier



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